

1 The curve **C** has equation $y = x^2 + 3x - 3$

The line **L** has equation $y - 5x + 4 = 0$

Show, algebraically, that **C** and **L** have exactly one point in common.

(Total for Question 1 is 4 marks)

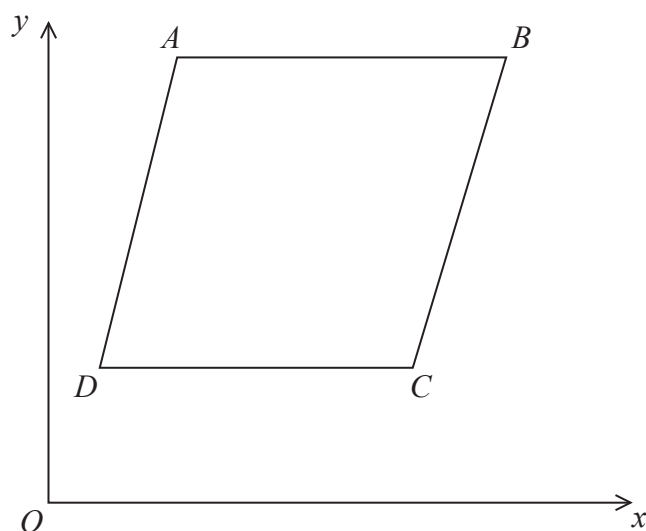
- 2** The point P has coordinates $(3, 4)$
The point Q has coordinates (a, b)

A line perpendicular to PQ is given by the equation $3x + 2y = 7$

Find an expression for b in terms of a .

(Total for Question 2 is 5 marks)

3



$ABCD$ is a rhombus.

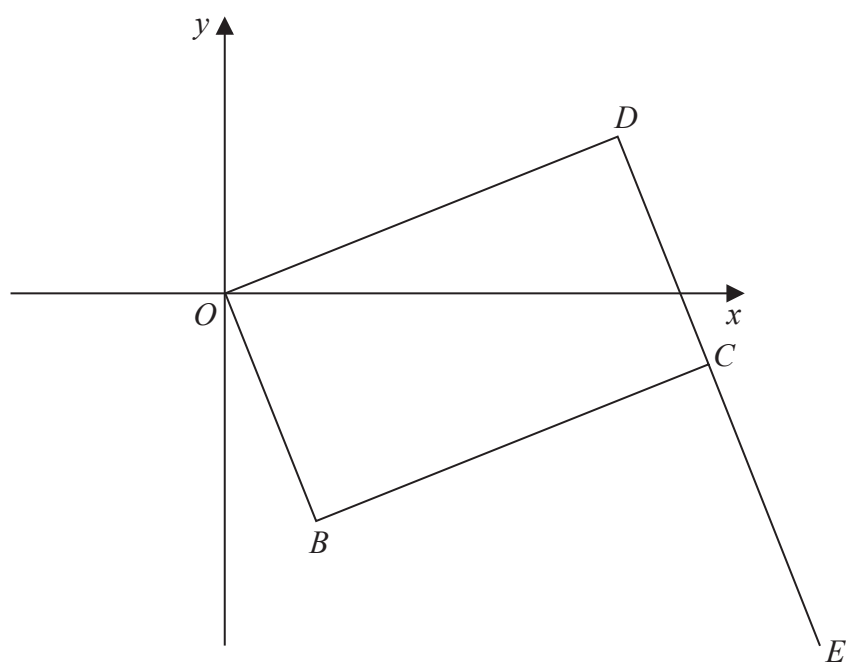
The coordinates of A are $(5, 11)$

The equation of the diagonal DB is $y = \frac{1}{2}x + 6$

Find an equation of the diagonal AC .

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(Total for Question 3 is 4 marks)

- 4 $OBCD$ is a rectangle.
 DCE is a straight line.



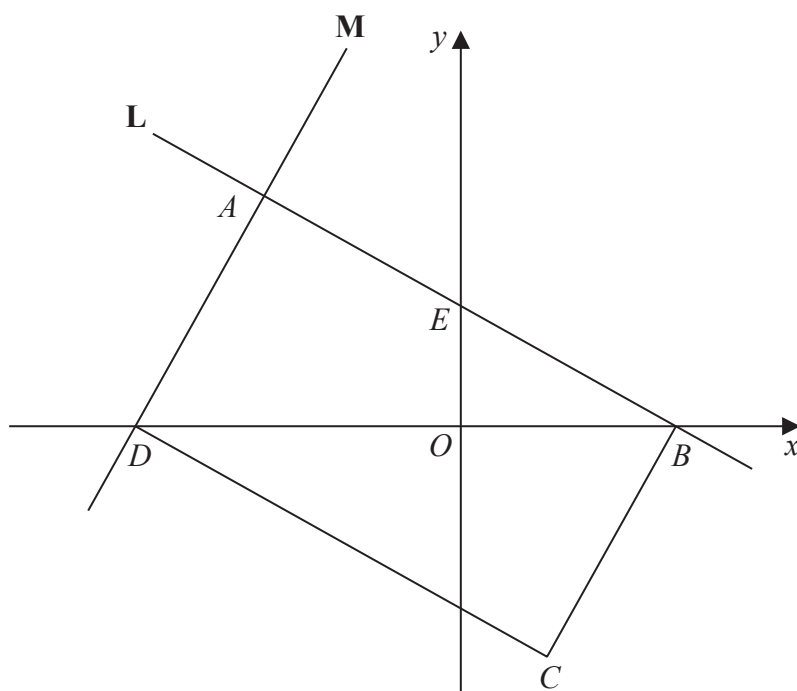
B has coordinates $(2, -4)$
 E has coordinates $(12, -6.5)$

Work out the coordinates of D .
You must show all your working.

(..... ,)

(Total for Question 4 is 5 marks)

5



$ABCD$ is a rectangle.

A , E and B are points on the straight line **L** with equation $x + 2y = 12$

A and D are points on the straight line **M**.

$$AE = EB$$

Find an equation for **M**.

(Total for Question 5 is 4 marks)

6 Line **L** has equation $4y - 6x = 33$

Line **M** goes through the point $A(5, 6)$ and the point $B(-4, k)$

L is perpendicular to **M**.

Work out the value of k .

(Total for Question 6 is 4 marks)

- 7 The straight line L_1 passes through the points with coordinates $(4, 6)$ and $(12, 2)$
The straight line L_2 passes through the origin and has gradient -3

The lines L_1 and L_2 intersect at point P .

Find the coordinates of P .

(..... ,)

(Total for Question 7 is 4 marks)

- 8 Prove algebraically that the straight line with equation $x - 2y = 10$ is a tangent to the circle with equation $x^2 + y^2 = 20$

(Total for Question 8 is 5 marks)

9 The centre of a circle is the point with coordinates $(-1, 3)$

The point A with coordinates $(6, 8)$ lies on the circle.

Find an equation of the tangent to the circle at A .

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(Total for Question 9 is 4 marks)

10 A circle has equation $x^2 + y^2 = 12.25$

The point P lies on the circle.

The coordinates of P are $(2.1, 2.8)$

The line L is the tangent to the circle at point P .

Find an equation of L .

Give your answer in the form $ax + by = c$, where a , b and c are integers.

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(Total for Question 10 is 4 marks)

- 11** The line l is a tangent to the circle $x^2 + y^2 = 40$ at the point A .
 A is the point $(2, 6)$.

The line l crosses the x -axis at the point P .

Work out the area of triangle OAP .

(Total for Question 11 is 5 marks)

12 **C** is a circle with centre $(0, 0)$

L is a straight line.

The circle **C** and the line **L** intersect at the points P and Q .

The coordinates of P are $(5, 10)$

The x coordinate of Q is -2

L has a positive gradient and crosses the y -axis at the point $(0, k)$

Find the value of k .

$k = \dots\dots\dots$

(Total for Question 12 is 5 marks)

13 The straight line **L** has equation $3x + 2y = 17$

The point *A* has coordinates (0, 2)

The straight line **M** is perpendicular to **L** and passes through *A*.

Line **L** crosses the *y*-axis at the point *B*.

Lines **L** and **M** intersect at the point *C*.

Work out the area of triangle *ABC*.

You must show all your working.

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(Total for Question 13 is 5 marks)

- 14** $A(-2, 1)$, $B(6, 5)$ and $C(4, k)$ are the vertices of a right-angled triangle ABC .
Angle ABC is the right angle.

Find an equation of the line that passes through A and C .

Give your answer in the form $ay + bx = c$ where a , b and c are integers.

(Total for Question 14 is 5 marks)

15 The point P has coordinates $(-4, 5)$

The point Q has coordinates $(6, -6)$

The point R has coordinates $(k, k + 3)$

Given that angle PRQ is a right angle,

find the possible values of k .

You must show all your working.

(Total for Question 15 is 5 marks)